



m i n u t e
c r y p t i c
d e c r y p t e r



An Introduction



Minute Cryptic

Solve a clue with
a hidden meaning.

Today's clue

Neo-liberal produces writing with a lot of feeling?
(7)

play

1 December, 2025 · By Member: Abdul K



An Introduction



Minute Cryptic

- Crossword game that incorporates letterplay and wordplay to create a linguistic puzzle
- 3 Components of a puzzle: Fodder, Indicator, and Definition
- Various types of puzzles (and combinations of puzzles!)
 - Anagrams: Take some letters and scramble them to make a new word
 - Selectors: Picking specific letters from each word
 - Hiddens: Word is hidden amongst the clue
 - And several more! Substitutions, Reversals, Homophones, etc!

indicator
↓
Monkey with no cigar,
as nature intended (7)
↑ definition ← fodder
 ↑ answer length



Introduction



2 December, 2025

By Member: Col Dod 2



Sitting in class, I'm pleading for a piece of cake (6)



par

hints

check



Introduction



2 December, 2025

By Member: Col Dod 2



Sitting in class, I'm pleading for a piece of cake (6)



This clue's indicator is 'Sitting in', and it's a hidden word indicator. 'Sitting in' a neighbouring string of fodder is our answer (hiding).



par

hints

check



Introduction



2 December, 2025

By Member: Col Dod 2



Sitting in class, I'm pleading for a piece of cake (6)

--	--	--	--	--	--



par

hints

check



This clue's fodder is 'class, I'm pleading'.
We'll use these letters (or some of them) to
construct our answer, as guided by our
indicator(s).



Introduction



2 December, 2025

By Member: Col Dod 2



Sitting in class, I'm pleading for a piece of cake (6)

--	--	--	--	--	--



par

hints

check



This clue's fodder is 'class, I'm pleading'.
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Introduction



2 December, 2025

By Member: Col Dod 2



Sitting in class, I'm pleading for a piece of cake (6)

--	--	--	--	--	--



par

hints

check



Our definition here is 'a piece of cake'. Our answer will be a way of saying that.



Introduction



2 December, 2025

By Member: Col Dod 2



Sitting in class, I'm pleading for a piece of cake (6)

s i m p l e



par

hints

check



Objective



- **To create a program to solve a minute cryptic clue**
 - First, by classifying the puzzle type given the indicator
 - Then, producing potential answers via wordplay
 - Eventually, using definitions, producing a most likely answer





- **Gather Minute Cryptic Data**
 - Compile a training set of ~30 acceptable puzzles
 - Compile a test set of ~10 acceptable puzzles
- **Create a Classifier:**
 - Classify what type of puzzle we are attempting to solve using multinomial logistic regression (Sci-kit)
 - Add features like Cosine Similarity, Length of Clues, and so on
- **Create a Solver:**
 - For every type of puzzle, we designed an algorithm to systematically solve for the correct solution
 - N-grams, Word2Vec, Dictionaries...
- **Assign probabilities to potential answers**
 - Using cosine similarity between vectors for each candidate and the definition (average vector in the case of multi-word definitions)



Data Gathering



Anagrams	Deliver extremely popular puzzling video	puzzling	video	Deliver	7
Selectors	Even nerdy guys can be badass	even	nerdy guys	can be badass	4
Anagrams	Changes actress!?	changes	actress	changes actress	7
Hiddens	Provide a lipstick sample for model	sample	Provide a lipstick	model	5
Selectors	Sister's half brothers	half	brothers	sister's	4
Selectors	Intermittent pain in body to irritate	intermittent	pain in body	to irritate	5
Hiddens	Temporary contract in government sector	sector	contract in government	temporary	6
Selectors	Dank memes on Instagram sent to principals	Principals	memes on Instagram sent to	Dank	5
Hiddens	Cinema illegally hides message on a screen	hides	Cinema illegally	message on a screen	5
Selectors	Go thirds in pet turkey	thirds	pet turkey	go	3
Anagrams	De Niro's acting decreased with Heat	acting	De Niro	decreased with Heat	6
Anagrams	Minute Cryptic finally solved (not against the clock)	cryptic	minute	not against the clock	7
Selectors	Original amigos each only halfway through paperwork	halfway	original amigos	paperwork	7
Hiddens	Inside Stephanie's revealing skirt	revealing	inside stephanie's	skirt	8
Reversals	Eye doctor orders bifocals and heads back home	Back	Eye doctor orders bifocals and	back home	5
Selectors	Eye doctor orders bifocals and heads back home	heads	Eye doctor orders bifocals and	back home	5
Hiddens	Stab piercing rogue's shield	peircing	rogue's shield	stab	5
Anagrams	Puzzle book's debut released by beaming eccentric	puzzle	beaming	puzzle	6
Reversals	Last five cappuccinos sent back in waves?	back	cappuccinos	waves	5
Hiddens	Last five cappuccinos sent back in waves?	last five	cappuccinos	waves	5
Anagrams	Love us madly? Grab 1000 books!	madly	love us	books	7
Selectors	Feel mature after clearing elm property	after clearing elm	feel mature	property	7
Hiddens	Perhaps backpacking close by Amsterdam?	backpacking	close by Amsterdam	perhaps	5



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Unlocking kindness, insight and humility, starts with reflection!?(5)



the prolific James and Ollie.

Minute Cryptic Clue 517: Unlocking kindness, insight and humility, starts with reflection!?(5)



Minute Cryptic
33K subscribers

Subscribe

576



Share



Ask



Download





Data Gathering



Anagrams		Deliver extremely popular puzzling video		puzzling	video	Deliver		7
Selectors		Even nearly clue can be hedged		even	nearly clue	can be hedged		4
Date	Minute Cryptic Category	Clue	Indicator (each word, we will sum them later)		Fodders (each word new line)		Definition (each word new line)	Number of Letters
April 1, 2025		Did butterfly finally flutter away from insect group (4)						
April 2, 2025		isappea_fr_m cam_us! (4, 3)						
April 3, 2025	Anagram	De Niro's acting decreased with Heat (6)	acting		De Niro		decreased with Heat	6
April 4, 2025		Devours wool & furs, spits out our feathers (5, 4)						
April 5, 2025		Does weed. Lacks leadership. Turns 18+ (4)						
April 6, 2025	Anagram +one	Minute Cryptic finally solved (not against the clock) (7)	cryptic		minute		not against the clock	7
April 7, 2025	selector	Original amigos each only halfway through paperwork (7)	halfway		original amigos		paperwork	7
April 8, 2025	wordplay	Epsilon after O... mega? (4)						
April 9, 2025	indicator is fodder...	Harvest turnip beside banks of creek? (4)	turnip					
April 10, 2025	Anagram, definition?	Remains hopeful heading into rough seas (5)	rough		seas, hopeful		remains	5
April 11, 2025	crazy actually	One silver piercing belonging to me. Fancy! (7)						
April 12, 2025	whole clue was definition a	Master of ceremonies holds one! (3)						
April 13, 2025	hidden	Inside Stephanie's revealing skirt (8)	revealing		inside stephanie's		skirt	8
April 14, 2025		500th out-take from funny Brad Pitt cameo (3,4)						
April 15, 2025	weird workplay	Chances of winning big in bridge? (4)						
April 16, 2025	definitional	Can't stand around bomb site and be indecisive (8)	bomb site					
April 17, 2025	reverse	Eye doctor orders bifocals and heads back home (5)	Back		Eye doctor orders bifocals and		back home	5
April 17, 2025	selectors	Eye doctor orders bifocals and heads back home (5)	heads		Eye doctor orders bifocals and		back home	5
April 18, 2025		Corrupted Macs with fake antivirus protection (4, 4)						
April 19, 2025	hidden	Stab piercing rogue's shield (5)	peircing		rogue's shield		stab	5
April 20, 2025	definitional	Maybe love to hate ally of Caesar, J. (7)						
April 21, 2025	definitional	Owen's stomach turned after dough-nut (6)						
April 22, 2025	Anagram +one	Puzzle book's debut released by beaming eccentric	puzzle		beaming		puzzle	6
April 23, 2025	Definitonal	Cockney's house with "Final Inspection" sign (4)						
April 24, 2025	reverse	Last five cappuccinos sent back in waves? (5)	back		cappuccinos		waves	5
April 24, 2025	hidden	Last five cappuccinos sent back in waves? (5)	last five		cappuccinos		waves	5
April 25, 2025	anagram + one	Love us madly? Grab 1000 books! (7)	madly		love us		books	7
April 26, 2025	definitional	drUg-deaLing casE! (7)						
Selectors		Feel mature after clearing elm property		33K subscribers		576		7
Hiddens		Perhaps backpacking close by Amsterdam?		backpacking		close by Amsterdam		5



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Selectors		Feel mature after clearing elm property	minute cryptic	Subscribe	576	7
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Indicator Cracker



anagrams



selectors



hiddens





Indicator Cracker



```
=== Test a New Unseen Clue ===  
Enter the clue: Provide a lipstick sample for model  
Enter the indicator: sample  
Enter the fodder words: Provide a lipstick  
Enter the definition: model  
Enter the solution length: 5
```

```
=== Prediction Result ===  
Predicted Category: Selectors
```

```
Probabilities:  
  Anagrams: 0.0142  
  Hiddens: 0.4085  
  Selectors: 0.5773
```



Indicator Cracker



- Some pretty good results...

```
# feature matrix
x = df[[
    "length",
    "fodder_length",
    "fodder_word_count",
    "w2v_anagram",
    "w2v_hidden",
    "w2v_selector"
]].values.astype(float)
```

```
=== Classification Report ===
```

	precision	recall	f1-score	support
Anagrams	1.00	0.67	0.80	3
Hiddens	1.00	0.67	0.80	3
Selectors	0.60	1.00	0.75	3
accuracy			0.78	9
macro avg	0.87	0.78	0.78	9
weighted avg	0.87	0.78	0.78	9



Indicator Cracker



- We tried adding other features...
 - But L2 complexity penalties stopped us in our tracks

```
=== Classification Report ===
              precision    recall  f1-score   support

   Anagrams       1.00      0.67      0.80         3
    Hiddens       1.00      0.67      0.80         3
   Selectors       0.60      1.00      0.75         3

 accuracy              0.78         9
 macro avg       0.87      0.78      0.78         9
 weighted avg     0.87      0.78      0.78         9
```

	precision	recall	f1-score	support
Anagrams	1.00	0.67	0.80	3
Hiddens	0.60	1.00	0.75	3
Selectors	0.50	0.33	0.40	3
accuracy			0.67	9
macro avg	0.70	0.67	0.65	9
weighted avg	0.70	0.67	0.65	9



Algorithmic Solving!



```
=== Minute Cryptic Decrypter + Word2Vec Meaning Matcher ===
```

```
Enter the fodder: Provide a lipstick
```

```
Enter the answer length: 5
```

```
Enter category (anagram / hidden / selector): selector
```

```
Multiple candidate words: {'provi', 'stick', 'ideal'}
```



Anagrams



Step 1: Find all the anagrams of the fodder

Step 2: Filter the anagrams for “real” words

Step 3: find the “+1 variants” for each original anagram

Step 4: filter the variants

Step 5: combine the filtered lists and remove the original fodder



tea

tea
tae
eat
eta
ate
aet

tea
eat
eta
ate

tea
atea
btea
ctea
...
aetx
aety
aetz
...

tea
atea
btea
ctea
...
aetx
aety
aetz
...

bate
...
eat
...
zeta



Selectors



Step 1: Find all the different types of “indicators” for selectors

Step 2: Create a list of all possible words based on all the clues

Step 3: Filter the list through the 100k most common words

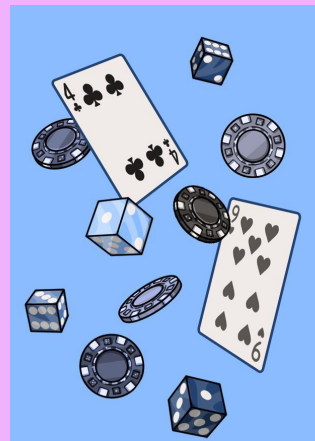
```
def first_letters(words):  
    return "".join(w[0] for w in words if len(w) > 0)  
  
def last_letters(words):  
    return "".join(w[-1] for w in words if len(w) > 0)  
  
def middle_letters(words):  
    return "".join(w[len(w)//2] for w in words if len(w) >= 3)
```

```
def nth_letters(chars, n):  
    return "".join(chars[0::n])  
  
def odd_letters(chars):  
    return "".join(chars[0::2])  
  
def even_letters(chars):  
    return "".join(chars[1::2])
```

half of a word

...

and so on



Intermittent pain in body
to irritate (5)

pibt
nidt
...
annoy

annoy



Hiddens



Step 1: “clean” the fodder using regex

Step 2: Find all n-grams of the fodder length

Step 3: filter among the most most frequent 100k English word

“Temporary contract in government sector” (6)

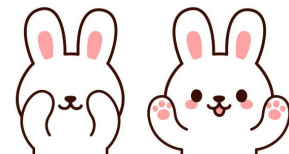
contract in government

"contract in government"

"contractingovernment"

{'acting','artnoc','cartno','contra',
'ctingo','emnrev','ernmen','evogni',
'Gnitca','govern','ingove','itcart','mn
revo','nemnre','ngover','nitcar',
'nrevog','ntact','ognitc','ontrac',
'overnm','ractin','revogn','rnment',
'tcartn','tingov','tnemn','tracti',
'vernme','vognit'}

PEEKABOO!



{'acting',
'contra',
'govern'}



Algorithmic Solving!



```
Multiple candidate words: {'provi', 'stick', 'ideal'}
```

```
Enter the DEFINITION part of the clue: model
```

```
=== Word2Vec Scoring ===
```

```
Definition: model
```

```
Best Match: ideal
```

```
ideal          0.6242
```

```
stick          0.3179
```

```
provi          0.0000
```

```
Final Answer: ideal
```



Potential Next Steps



- Account for **more puzzle types**
 - Definitional substitutions
 - Multi-word answers
 - Multiple indicators
- Make the program figure out what the indicator, fodder, and definitions are on its own (**part of speech tagging** maybe)
- **Hyperparameter optimization** in the machine learning (lambda smoothing)
- Have a **larger training set**



substitutions



containers



deletions



homophones





**Thank You
For
Listening!**
Questions?